

RAD-SMART

The right patient, on the right machine, at a predictable time.

Radiotherapy AI-assisted Dynamic Scheduling, Machine Allocation and Resource Tracking



One planned treatment day: each block is a patient's session (routine, new start, complex, blood irradiation)

Doctor partner: Dr Akshay Dinesan, Manipal · Technology team: [names]

Assistive, not diagnostic

THE PROBLEM

Every weekday, 80 patients wait for one machine

Appointment times are given by hand, without knowing how many machine-minutes each patient needs. Too many people are told to come at once, and they wait, every day of their course.

70–90

patients treated on one machine
every working day

5–7 weeks

of daily visits for a curative course,
usually with a caregiver

69%

of India's 554 radiotherapy
facilities have only one machine

Allocation under hard rules and soft priorities

Hard rules: never broken

- New starts finish before 5 PM
- SBRT, SRT, TBI and CSI inside the senior-staff window, 10 AM – 5 PM
- Blood irradiation slot, 1:30 – 2 PM
- Breast boards and ABC shared with the CT simulator
- Versa HD and Halcyon capabilities
- Capacity held for same-day urgent starts

Soft priorities: traded off

- Elderly and public-transport patients earlier
- Inpatient, hospice and dormitory time windows
- Paying-patient preference, as the department sets it
- Complex cases in the 12:00 – 1:30 protected block
- As little overtime as possible

And every day brings events: machine faults, overruns, no-shows and urgent add-ons, each of which should trigger a re-plan.

FIVE APPROACHES

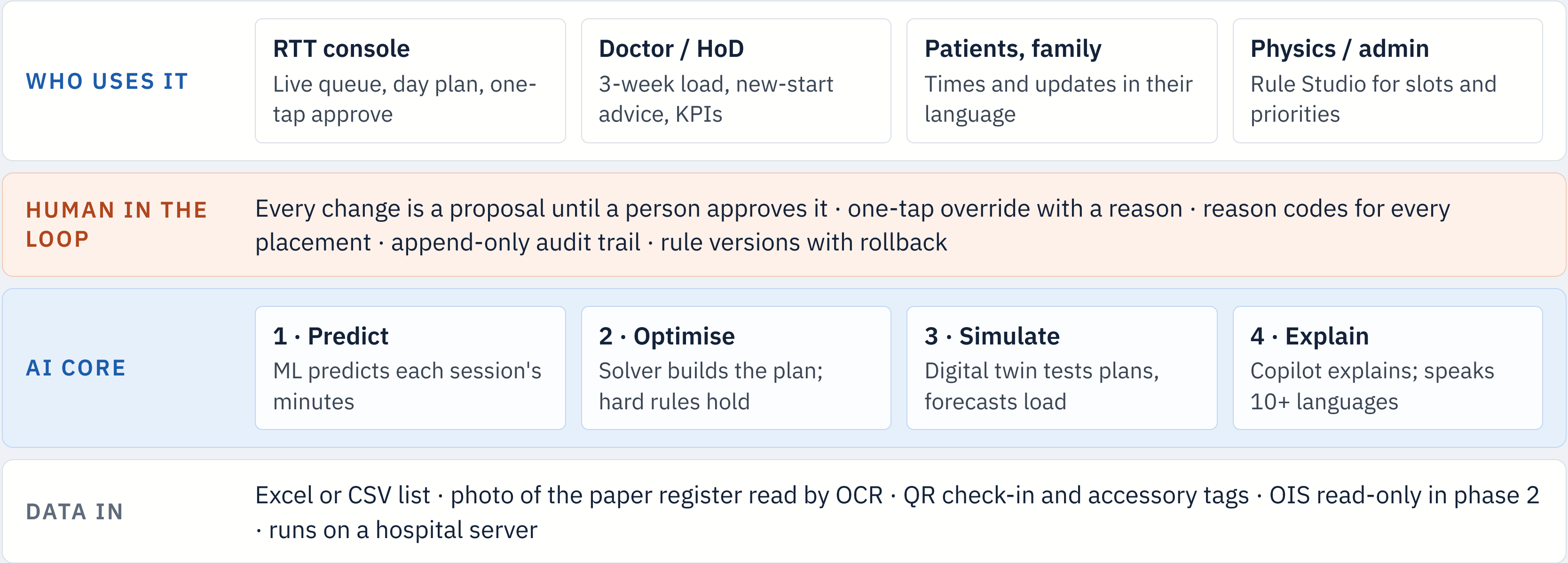
We compared five ways to build it

Approach	Strength	Weakness	Score
Rule-based slot templates	Simple, transparent	Can't absorb variation or faults	3.10
LLM-only agent	Conversational	No guarantee on hard rules	2.60
Static optimisation	Guarantees hard rules	No learning or live recovery	3.60
Deep reinforcement learning	Adaptive in principle	Opaque; needs huge data	2.45
Hybrid (selected): predict, optimise, simulate, explain	Best of each technique	More parts, built in layers	4.85

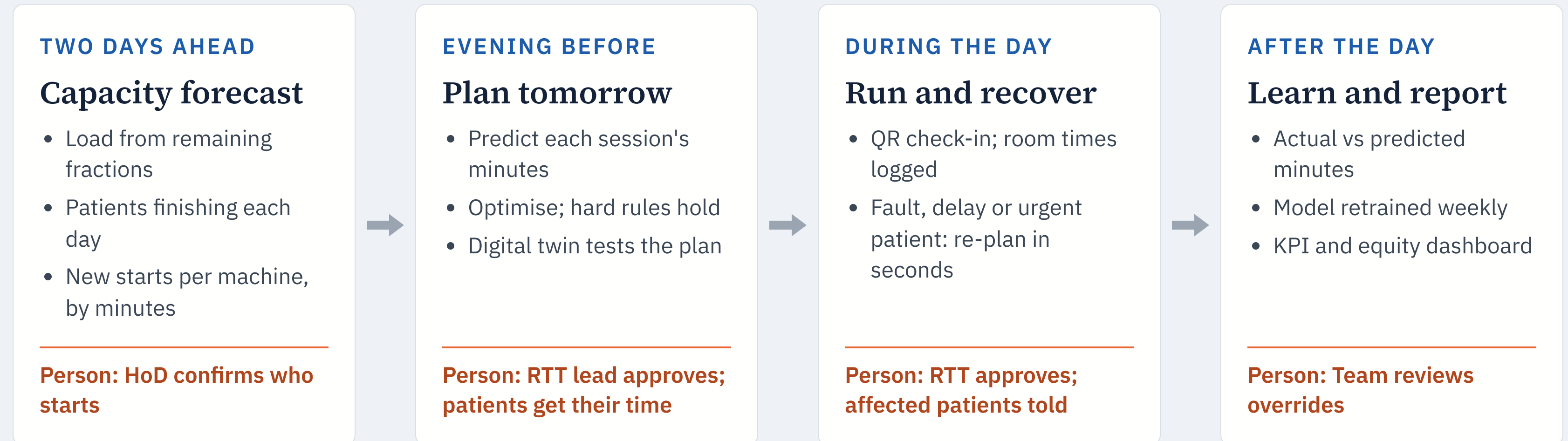
Weighted score out of 5: safety and scope 20%, impact 20%, real-time 15%, feasibility 15%, explainability 10%, ease and languages 10%, configurability 10%.

THE SOLUTION

How RAD-SMART works



Automation proposes; people decide



Every override and actual timing becomes training data, so plans improve each week.

AI where each technique is strongest

PREDICT · MACHINE LEARNING

Minutes, not head-counts

Quantile gradient boosting gives typical (P50) and cautious (P80) minutes per session.

PoC: 1.9 min error vs 2.4 for averages

OPTIMISE · OPERATIONS RESEARCH

Hard rules guaranteed

OR-Tools CP-SAT (HiGHS in the PoC) builds and repairs the plan; hard rules cannot break.

PoC: proven optimum in 2–8 s

SIMULATE · DIGITAL TWIN

Tested before it is used

A Monte Carlo day model stress-tests plans and forecasts three weeks of load.

PoC: 300 simulated days per policy

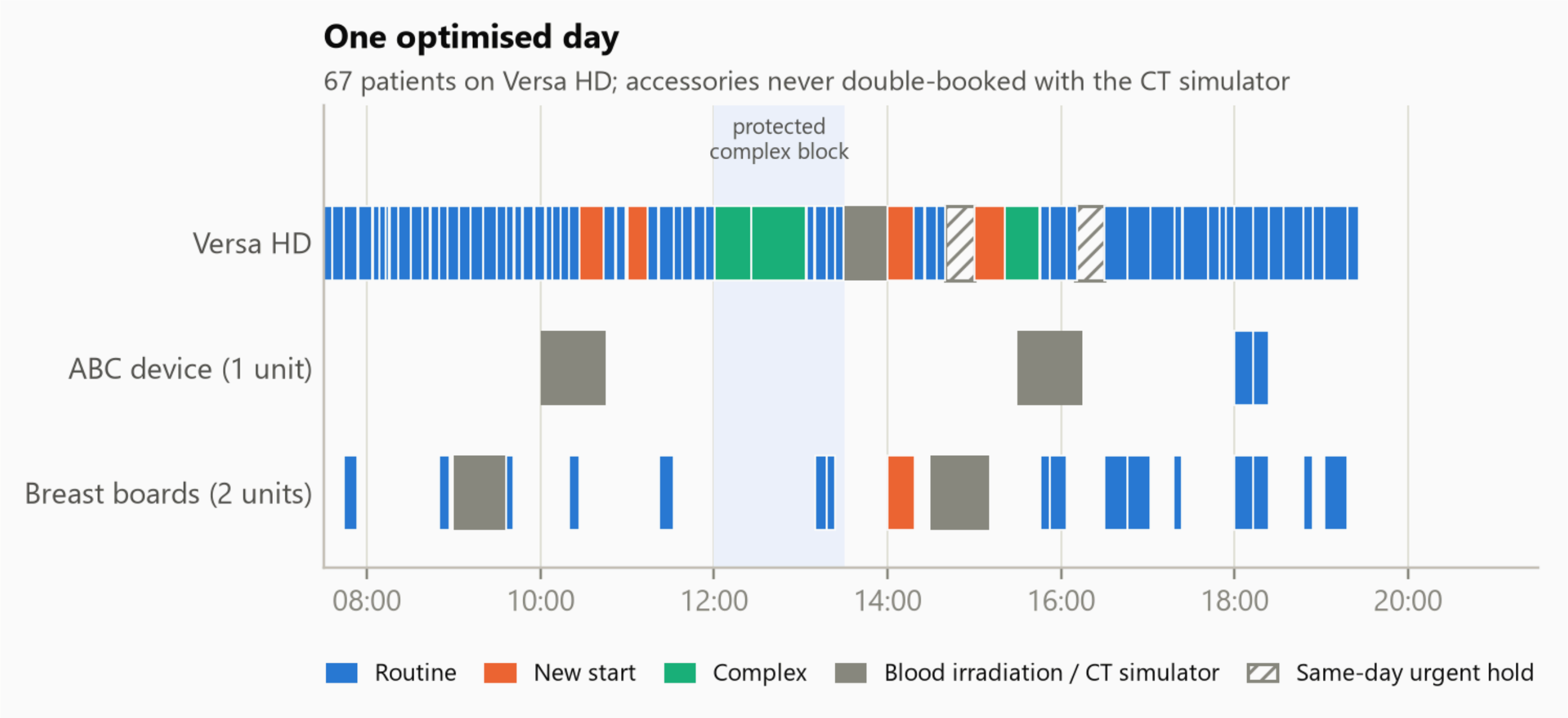
EXPLAIN · SARVAM AI

Plain language, 10+ languages

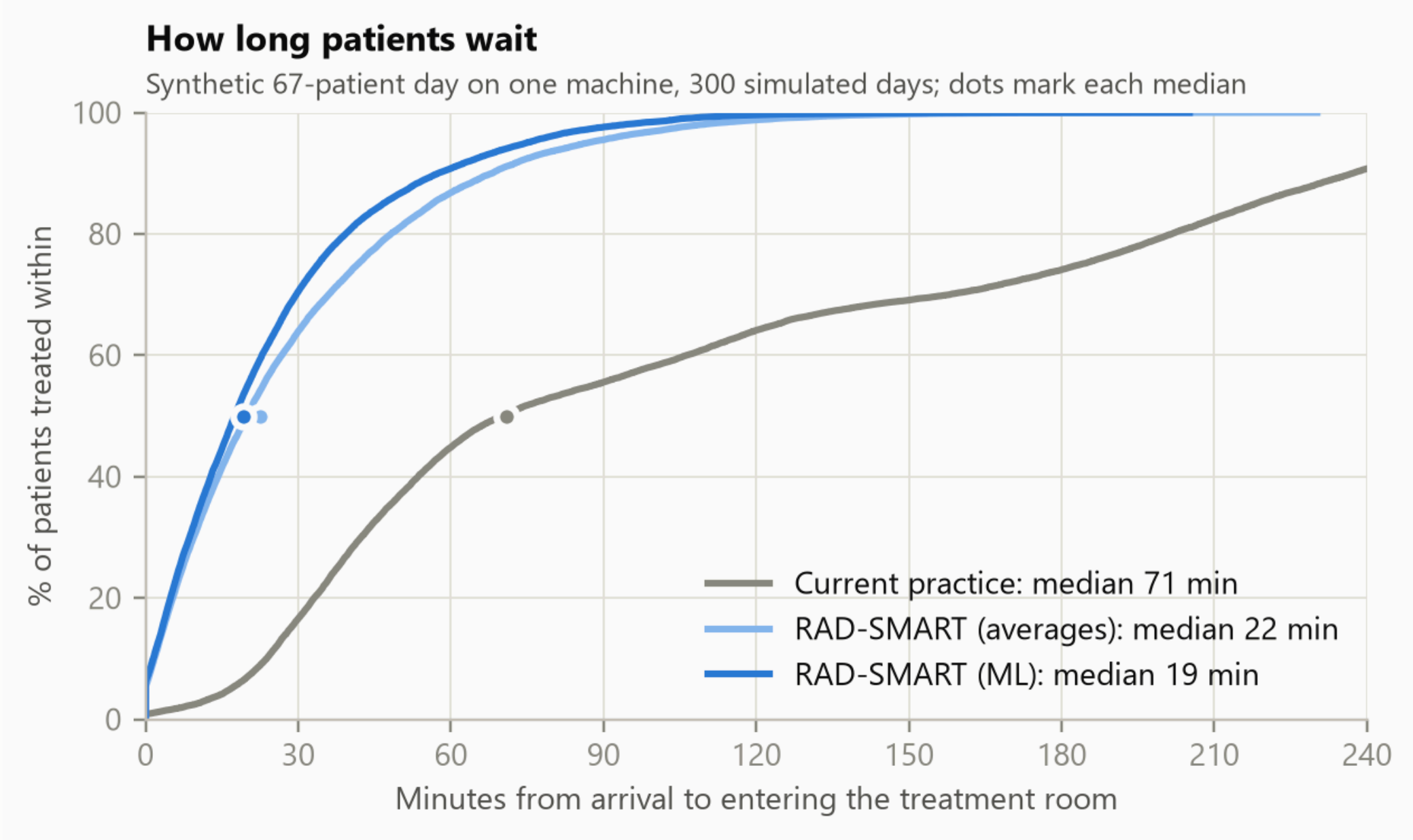
Sarvam-105B explains decisions and edits rules; Bulbul, Saaras and Translate reach patients.

The LLM never schedules

One optimised day, with every rule held



Waits fall by about three-quarters



71 → 19 min

median wait, arrival to treatment room

227 → 52 min

90th-percentile wait

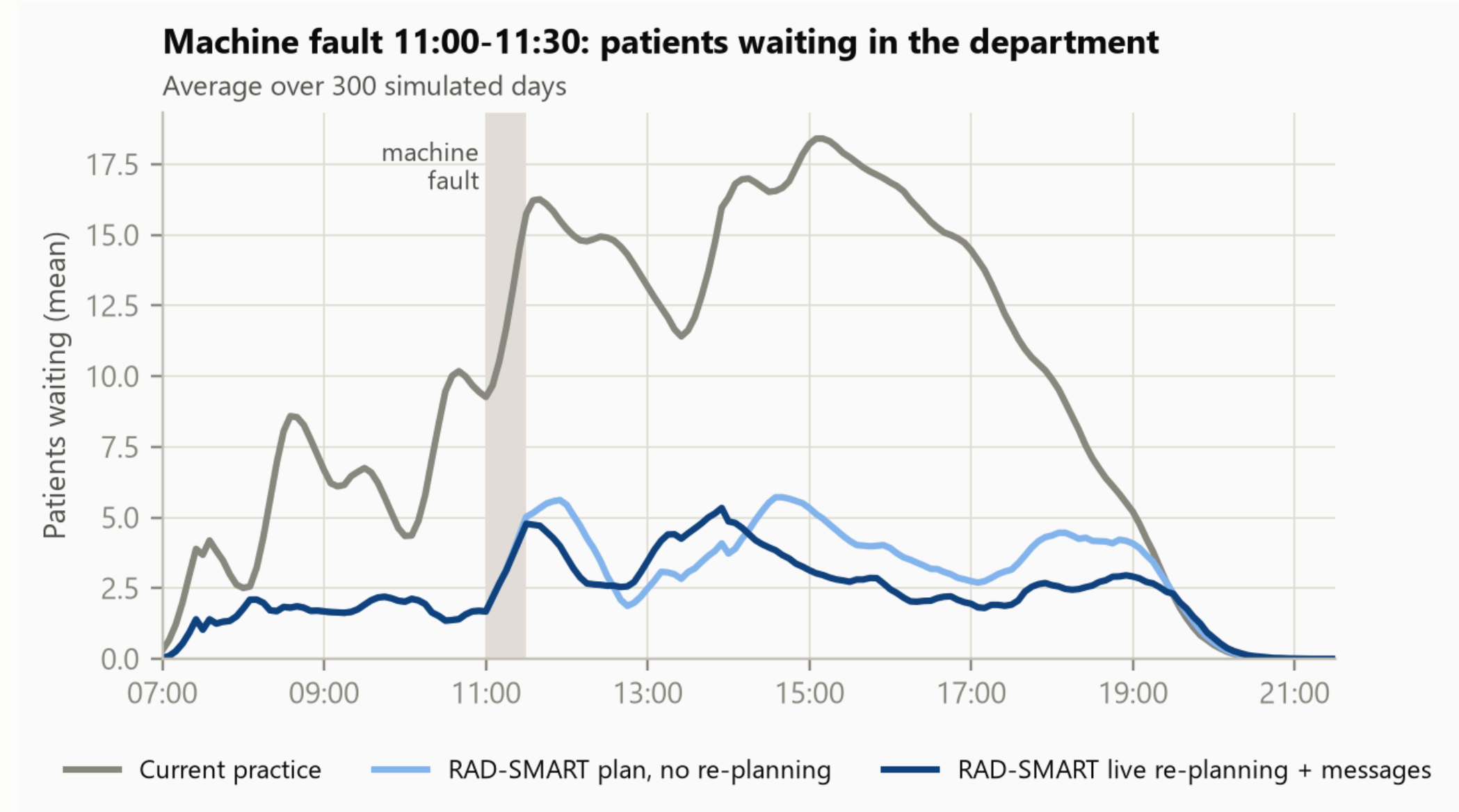
16% → 62%

treated within ±15 min of their time

No extra overtime

and no new start after 5 PM

It recovers when the machine fails



4.0 s

to re-plan the rest of the day after a 30-minute fault

38 moved, 17 told

only patients with 45+ minutes' notice are messaged

93 → 19 min

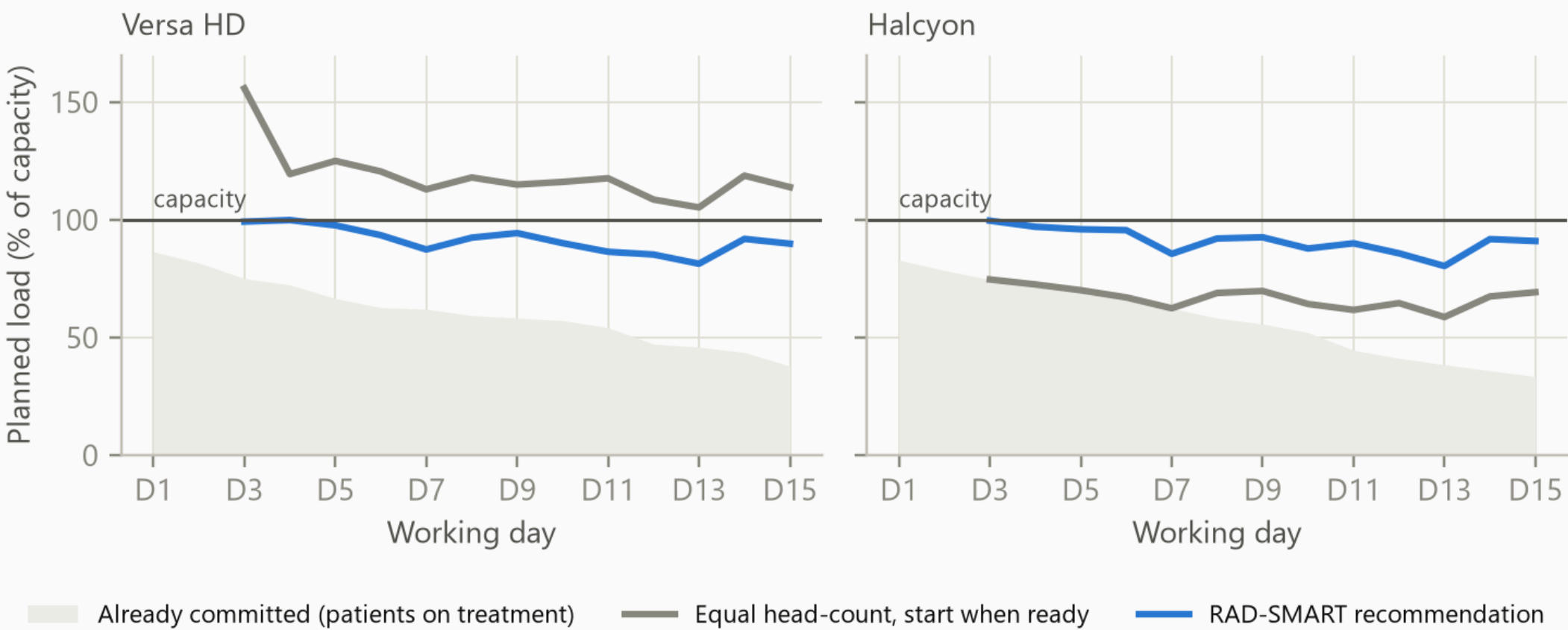
median wait, versus current practice

0

new starts after 5 PM or complex cases deferred

New starts balanced by minutes, not head-count

Three-week machine-load forecast with new-start recommendations



156% → 100%

peak planned load on the Versa HD

13 → 0

decision days over capacity

0 of 84

new patients still waiting after three weeks

2 days ahead

advice per machine, as the department asked

Safe by design

Proposal until approved

RTTs approve every plan and re-plan, and override in one tap with a reason.

Rules re-checked

An independent checker verifies every hard rule before a plan is published.

Clinicians decide

No diagnoses, doses, images or notes are read. Urgency and eligibility come from clinicians.

No identifiers to AI

Models see pseudonymous IDs; names and numbers are added on the hospital server.

Full audit trail

Every plan, override, rule version, model version and message is logged.

Works where the clinic is

Excel, paper via OCR, QR and WhatsApp; runs on the LAN with a printed fallback.

PILOT

90 days on the Versa HD, one primary KPI

Days 1–14

Baseline

Timestamps only;
scheduling
unchanged

Days 15–35

Shadow mode

RAD-SMART plans nightly; RTTs
compare

Days 36–90

Live

RTT lead approves each plan; patients get times and live updates; model retrained weekly

PRIMARY KPI

$\geq 40\%$ lower

median wait from arrival to linac entry, against a measured two-week baseline

ALSO TRACKED

- At least 50% of sessions within ± 15 min
- 100% of new starts finished before 5 PM
- Overtime not above baseline
- Throughput not lower; no patient group worse off

From prototype to pilot

Team

- Dr Akshay Dinesan, Manipal: doctor partner and problem owner
- [Technical lead]
- [Optimisation and ML engineer]
- [Full-stack and messaging engineer]

Roadmap

- 5 Oct – 8 Nov: build sprint
- 28 Nov: finale, IIT Bombay
- Nov – Jan: approvals
- Q1 2027: 90-day pilot
- Then: Halcyon, OIS link, NCG centres

Our ask

- A pilot slot with mentor support
- Sarvam API credits for voice and translation
- Introductions to NCG centres for scale-up

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